

Maha Sajid

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EDUCATION

Virginia Tech

Expected May 2028

PhD in Computer Science

Research Focus: Human-Computer Interaction, Collaborative Mixed Reality (MR), Team Cognition

Lahore University of Management Sciences (LUMS)

Aug 2018 – May 2022

BS in Computer Science

Minor in Psychology · Graduated with Distinction

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug 2023 – Present

Virginia Tech

Blacksburg, VA

Advisors: Dr. Brendan David-John and Dr. Bo Ji

Collective Situational Awareness in Asymmetric MR:

- Led an end-to-end investigation of how communication modality shapes collective situational awareness in asymmetric MR, designing and implementing a networked HoloLens 2 experiment, refining the protocol through a **12-participant pilot**, and conducting an IRB-approved, counterbalanced study with **48 participants across 24 dyads**.
- Extended SAGAT-based collective situational awareness assessment by **introducing two team-level measures**: information coverage and accurate alignment, showing that aggregate scores can obscure meaningful differences in how knowledge is distributed and shared across teams.
- Built **multimodal analysis pipelines** integrating gaze, speech, situational awareness, workload, and performance data, deriving task-space gaze and verbal-coordination measures to characterize how teams coordinated across conditions.
- Found that **Gaze+Voice improved task performance** and **shared awareness**, while Gaze-only produced the strongest spatial gaze convergence, showing that performance, co-attention, and collective awareness capture distinct aspects of collaboration. Manuscript is under review at **IEEE VR 2027**.

Security Attacks in Remote Collaborative MR:

- Designed and executed a mixed-methods security study on how users recognize and respond to attacks in remote collaborative MR, iteratively refining the task, attack conditions, and study protocol across **two pilot studies with 16 participants** before conducting an IRB-approved main study with **20 participants across 10 dyads**.
- Engineered **four real-time security attack interventions** in a networked Unity-based HoloLens 2 platform, including **a novel click-redirection mechanism** that manipulated 3D object interactions and **MR-specific occlusion attacks** that exploited invisible spatial elements.
- Triangulated first-person recordings, behavioral observations, questionnaires, and semi-structured interviews to uncover gaps between users' perceived and actual security behavior; **54% reported no attack during spatial occlusion**, while **67% failed to report the mitigation strategy they had actually used**.
- Translated findings on attack misattribution and mitigation into **design recommendations** for immersive security indicators, interaction transparency, and user training; resulted in a **first-author IEEE VR 2025 paper** featured in *Computerworld*.

Bystander Privacy Indicators in Extended Reality (XR):

- Collaborated on an iterative mixed-methods study of privacy indicators for situationally impaired XR bystanders, using four **focus groups** with 8 participants to **develop five visual and multimodal concepts** followed by a 7-participant evaluation of usability and contextual usefulness.
- Analyzed qualitative and quantitative feedback across indicator designs, finding that visual-only cues were less effective under situational impairment while multimodal indicators were preferred in privacy-sensitive scenarios; the work resulted in an **IEEE ISMAR Adjunct 2025 publication**.

Undergraduate Research Assistant

Jan 2022 – Feb 2023

Lahore University of Management Sciences (LUMS)

Lahore, Pakistan

Advisors: Dr. Fareed Zaffar and Dr. Murtaza Taj

- Achieved **92% accuracy** in identifying constructed areas by developing a satellite-imagery analysis pipeline for detecting land-use change and potentially unrecorded development, supporting prioritization of labor-intensive property-tax surveys.
- Developed a **GIS-integrated decision-support platform** that transformed machine-learning outputs into spatial and temporal construction comparisons, enabling users to inspect newly detected construction across locations and years.

- Integrated remote-sensing outputs into a government property-verification workflow for flagging inconsistencies between GIS and taxation records and directing field-survey resources; the work contributed to the **International Growth Centre technical report** *Cross Verification of Revenue Using Remote Sensing Data*.

SELECTED PUBLICATIONS

- **Sajid, M.**, Bukhari, S. I. M. S., Ji, B., & David-John, B. (2025). *“Just stop doing everything for now!”: Understanding Security Attacks in Remote Collaborative Mixed Reality*. **IEEE VR 2025**. DOI: 10.1109/VR59515.2025.00085.
- Bukhari, S. I. M. S., **Sajid, M.**, Ji, B., & David-John, B. (2025). *Rethinking Privacy Indicators in Extended Reality: Multimodal Design for Situationally Impaired Bystanders*. **IEEE ISMAR Adjunct 2025**. DOI: 10.1109/ISMAR-Adjunct68609.2025.00059.

INDUSTRY EXPERIENCE

Software Engineer

Feb 2023 – Jul 2023

Senarios

Lahore, Pakistan

- Delivered a cross-platform mobile data-sharing product from design through release, implementing real-time contact synchronization, decentralized functionality, and technical specifications for a Google Play deployment used by active users.
- Partnered with **3+ clients** to translate Web3 data-integrity requirements into implementation plans and coordinate feature delivery across engineering and product stakeholders in an Agile environment.

Software Engineer

Jun 2022 – Jan 2023

PakWheels

Lahore, Pakistan

- Independently led the integration of payment gateways and supporting checkout workflows for an automotive marketplace, owning the feature from implementation through production deployment.
- Developed a **Python/XGBoost vehicle-pricing model** using used-car listing attributes to generate data-driven price recommendations for marketplace users.

RESEARCH & TECHNICAL SKILLS

Research Methods: Experimental design, human-subjects research, mixed-methods research, controlled and repeated-measures studies, pilot studies, focus groups, semi-structured interviews, behavioral observation, usability evaluation

Quantitative & Qualitative Analysis: ANOVA, Friedman and Wilcoxon tests, bootstrapping, effect sizes, clustering, eye tracking, gaze-object mapping, joint attention, gaze entropy, speech transcription and analysis, multimodal data synchronization, thematic analysis, qualitative coding

XR Prototyping & Experimentation: Unity, OpenXR, Microsoft HoloLens 2, Galaxy XR, Android XR, MRTK, XR Interaction Toolkit, networked multi-user XR, Mirror, Unity Netcode, Photon Fusion, Figma

Programming & Data Tools: Python, C#, C++, JavaScript, Node.js, React, Git, Pandas, NumPy, SQL, Ruby

TEACHING EXPERIENCE

Graduate Teaching Assistant

Dec 2024 – Dec 2025

Virginia Tech

- Courses: CS 1054: Intro to Programming in Java, CS/ECE 5590: System and Software Security, CS 5024: Ethics and Professionalism in Computer Science

Undergraduate Teaching Assistant

Aug 2022 – May 2023

LUMS

- Courses: CS 360: Software Engineering, CS 5310: Computer Vision Fundamentals

HONORS, LEADERSHIP & SERVICE

- **Awards & Fellowships:** John Lee Pratt Fellowship (2026–2027) · Two Commonwealth Cyber Initiative (CCI) Cyber Innovation Scholar Awards (\$3,500, 2024–2026). · TAPIA scholarship (2026)
- **Peer Review & Professional Service:** Reviewer, IEEE VR (2026) and Human-Centric Intelligent Systems (2026) · Organizing Committee, iXR Ideathon (2026) · Student Volunteer, IEEE ISMAR (2025).
- **Mentorship:** Mentored two Virginia Tech undergraduate researchers; Computing Research Association (CRA) UR2PhD Certified Graduate Mentor (2026).
- **Leadership & Honors:** Graduate Council Officer, Virginia Tech CS (2026–2027) · General Secretary, LUMS Data Science Society (2021–2022) · Dean’s Honor List, LUMS (2019–2022).